



PADJADJARAN STATISTICS OLYMPIAD 10.0
HIMPUNAN MAHASISWA STATISTIKA 2026
FAKULTAS MATEMATIKA DAN ILMU PENGETAHUAN ALAM
UNIVERSITAS PADJADJARAN.



INFOGRAPHIC DESCRIPTION – RASIO 10.0 2026

Where the Field Ends, the Problem Doesn't: Spatial-Panel Evidence That ASEAN's Food-System Pressure Is Shared

Can ASEAN's agri-food emissions be governed nation by nation? This infographic answers with a balanced panel of 44 Asia-Pacific countries, 1961–2024 (2,816 country-years), compiled from Our World in Data and the Global Carbon Budget (Jones et al., 2023; Ritchie et al., 2024), with population from the United Nations (2024). Its main message is that the region's food-system pressure is shared across borders, so unilateral policy captures only a fraction of what coordination delivers.

Four insights carry that message. First, scale is outrunning efficiency: ASEAN produced 22.78% of global land-use change (LUC) CO₂ in 2024 while holding 7.44% of total greenhouse gases, and forecasts to 2035 show agricultural methane rising 22.4% against only 5.3% population growth under walk-forward validation (Hyndman & Athanasopoulos, 2021). Second, K-Means clustering (silhouette = 0.534) groups seven of ten members into a "Frontier Seven" defined by land expansion, not input intensity. Third, the hotspot is tightening: global Moran's I rose from 0.255 in 1961 to 0.729 in 2024, with those seven states in persistent High-High clusters (Anselin, 1995). A Spatial Durbin Model ($\rho = 0.516$; AIC = 4752), decomposed following LeSage and Pace (2009), shows fertiliser N₂O carrying an indirect effect (+1.1046) more than twice its direct effect (+0.5011), so 68.8% of its impact crosses borders. Fourth, that spillover determines who must act: under a 20% nitrogen-efficiency scenario, five countries capture 81% of the achievable regional reduction.

These findings underpin GRAIN, a cross-border targeting framework run on the S.E.E.D cycle: Scan persistent hotspots, Engage regionally on nitrogen, Equip national and village methane programmes, and Disclose verified outcomes. Coordinated action cuts LUC CO₂ by 32.1% (335 Mt per year) against 10.0% (105 Mt) unilaterally 231 Mt forgone. Data reveal where the pressure gathers; cooperation decides whether it can be unwound.



PADJADJARAN STATISTICS OLYMPIAD 10.0
HIMPUNAN MAHASISWA STATISTIKA 2026
FAKULTAS MATEMATIKA DAN ILMU PENGETAHUAN ALAM
UNIVERSITAS PADJADJARAN.





PADJADJARAN STATISTICS OLYMPIAD 10.0
HIMPUNAN MAHASISWA STATISTIKA 2026
FAKULTAS MATEMATIKA DAN ILMU PENGETAHUAN ALAM
UNIVERSITAS PADJADJARAN.



REFERENCES

- Anselin, L. (1995). Local indicators of spatial association—LISA. *Geographical Analysis*, 27(2), 93–115.
- Box, G. E. P., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. (2015). *Time series analysis: Forecasting and control* (5th ed.). Wiley.
- Crippa, M., Solazzo, E., Guizzardi, D., Monforti-Ferrario, F., Tubiello, F. N., & Leip, A. (2021). Food systems are responsible for a third of global anthropogenic GHG emissions. *Nature Food*, 2, 198–209.
- Elhorst, J. P. (2014). *Spatial econometrics: From cross-sectional data to spatial panels*. Springer.
- Friedlingstein, P., O’Sullivan, M., Jones, M. W., Andrew, R. M., Bakker, D. C. E., Hauck, J., et al. (2023). Global Carbon Budget 2023. *Earth System Science Data*, 15(12), 5301–5369.
- Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice* (3rd ed.). OTexts.
- Hyndman, R. J., & Koehler, A. B. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4), 679–688.
- Jones, M. W., Peters, G. P., Gasser, T., Andrew, R. M., Schwingshackl, C., Gütschow, J., Houghton, R. A., Friedlingstein, P., Pongratz, J., & Le Quéré, C. (2023). National contributions to climate change due to historical emissions of carbon dioxide, methane, and nitrous oxide since 1850. *Scientific Data*, 10, Article 155.
- Klein Goldewijk, K., Beusen, A., Doelman, J., & Stehfest, E. (2017). Anthropogenic land use estimates for the Holocene — HYDE 3.2. *Earth System Science Data*, 9(2), 927–953.
- LeSage, J. P., & Pace, R. K. (2009). *Introduction to spatial econometrics*. CRC Press.
- MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations. *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, 1, 281–297.
- Moran, P. A. P. (1950). Notes on continuous stochastic phenomena. *Biometrika*, 37(1/2), 17–23.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–



PADJADJARAN STATISTICS OLYMPIAD 10.0
HIMPUNAN MAHASISWA STATISTIKA 2026
FAKULTAS MATEMATIKA DAN ILMU PENGETAHUAN ALAM
UNIVERSITAS PADJADJARAN.



2830.

Ritchie, H., Roser, M., & Rosado, P. (2024). CO2 and greenhouse gas emissions. Our World in Data. <https://ourworldindata.org/co2-and-greenhouse-gas-emissions>

Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65.

United Nations, Department of Economic and Social Affairs, Population Division. (2024). *World Population Prospects 2024*.



PADJADJARAN STATISTICS OLYMPIAD 10.0
HIMPUNAN MAHASISWA STATISTIKA 2026
FAKULTAS MATEMATIKA DAN ILMU PENGETAHUAN ALAM
UNIVERSITAS PADJADJARAN.



REFERENCES

Example:

Stevens, Izzie. (2008). Surgery for Trauma Patients. Surgeon Profession Journal, 3(1), 48-55.

Ashing-Giwa, K. T., Padilla, G., Tejero, J., Kraemer, J., Wright, K., Coscarelli, A., Clayton, S., Williams, I., & Hills, D. (2004). Understanding the breast cancer experience of women: A qualitative study of African American, Asian American, Latina and Caucasian cancer survivors. Psycho-Oncology, 13(6), 408-428. <https://doi.org/10.1002/pon.750>